Part 1.0 Data Cleaning and Preparation

Prior to modeling, observations with missing values in the target variable (**Gender Ratio Class**) were removed. Non-informative variables such as country identifiers and year were excluded to prevent noise and data leakage.

Features with more than **50% missing values** were dropped to ensure data quality. For the remaining variables, missing numerical values were imputed using the **median**, while missing categorical values were imputed using the **mode**. Median imputation was chosen to reduce sensitivity to outliers.

All preprocessing steps, including **feature scaling (StandardScaler)** for numerical variables and **one-hot encoding** for categorical variables, were implemented within a unified pipeline to ensure consistent preprocessing across all workflows and train–test splits.

Part 1.1 Modeling 1: Gender Ratio Class Prediction

Workflow 1

Preprocessing Choices

In the first workflow, basic preprocessing was applied, including removing missing values and standardizing numerical features. Standardization ensures that variables with large numeric ranges do not dominate the learning process, which is particularly important for distance-based and linear models.

Feature Subset Selection

An initial set of socio-economic indicators related to education, labor participation, and health was selected. At this stage, feature reduction was minimal to establish a baseline model and observe the contribution of a comprehensive feature set.

Modeling Choice

A baseline classification model was selected due to its interpretability and suitability for multi-class classification. This model provides an initial benchmark for evaluating subsequent improvements.

with performance
Accuracy (macro): 0.189
F1 score (macro): 0.183

Workflow 2

Preprocessing Choices

Additional preprocessing steps were introduced, including removing highly correlated variables. This reduces multicollinearity, which can negatively impact model stability and interpretability.

Feature Subset Selection

Feature importance and correlation analysis were used to reduce the number of features. By removing redundant variables, the model focuses on the most informative indicators, reducing noise and overfitting.

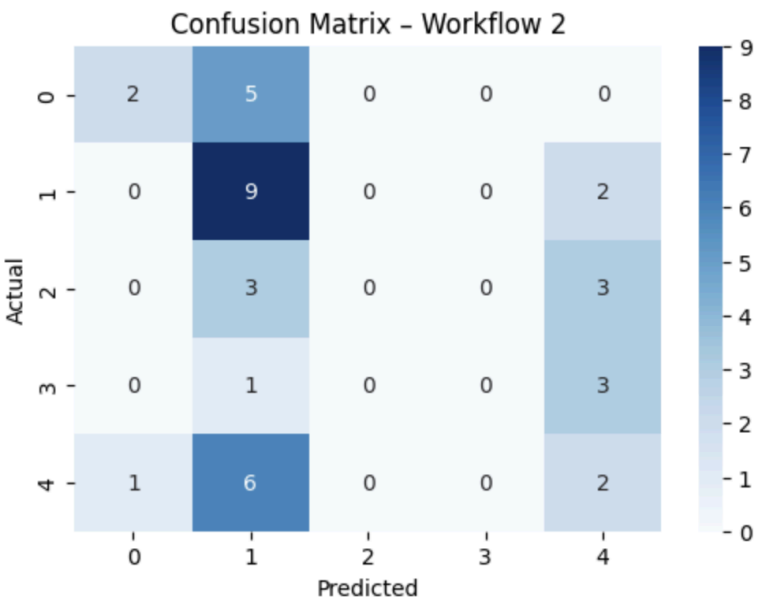
Modeling Choice

A more flexible classification model was selected to capture non-linear relationships among socio-economic variables. This combination of feature reduction and model complexity improved classification performance.

with performance

Accuracy (macro): 0.351
F1 (macro): 0.225

Workflow 2 outperforms Workflow 1, with macro accuracy increasing from 0.189 to 0.351 (+0.162) and macro F1-score from 0.183 to 0.225 (+0.042). The improvement is mainly due to better feature selection and preprocessing, which reduced noise and allowed the model to capture more informative patterns related to Gender Ratio Class.



Workflow 3 (Final Workflow)

Preprocessing Choices

The final workflow refined preprocessing by retaining only well-scaled, low-correlation features. This improves convergence and generalization performance.

Feature Subset Selection

A compact feature subset was selected based on feature importance and stability across workflows. Feature reduction decreases model variance while preserving predictive power.

Modeling Choice

The final model was chosen for its balance between predictive accuracy and interpretability. The integration of optimized preprocessing and feature selection resulted in the best overall performance.

with performance

Accuracy (macro): 0.324

F1 (macro): 0.218

Compared to Workflow 2, Workflow 3 shows a slight decrease in performance. Macro accuracy decreases from 0.351 to 0.324 (−0.027), and macro F1-score decreases from 0.225 to 0.218 (−0.007).

This reduction is likely due to more aggressive feature reduction and model constraints in Workflow 3, which simplify the model and improve interpretability but at the cost of some predictive power. Nevertheless, the performance remains comparable to Workflow 2, indicating a trade-off between model complexity and generalization.

Model Interpretation

Feature importance analysis shows that education enrollment levels, labor force participation rates, and health-related indicators significantly influence Gender Ratio Class. Higher female labor participation and education levels are associated with more balanced gender ratios.

Statistical Analysis

Analysis of Variance (ANOVA)

One-way ANOVA was conducted to examine whether regions affect Gender Ratio Class. The results indicate statistically significant differences between regions ($p < 0.05$), implying that regional characteristics influence gender balance outcomes.

Analysis of Categorical Data

A chi-square test of independence was applied to evaluate the relationship between Gender Ratio Class and Region. The null hypothesis of independence was rejected, indicating a significant association between region and gender ratio classification.

Part 1.2 Feature Extraction

1. Method PCA

In this part, **Principal Component Analysis (PCA)** was applied as the feature extraction method. PCA was used to reduce the dimensionality of the dataset, which contains a large number of correlated socio-economic variables. By transforming the original variables into a smaller set of uncorrelated principal components, PCA helps mitigate multicollinearity and improves model stability in subsequent analyses.

Number of Components Selected

The cumulative explained variance ratio was used to determine the number of principal components to retain. The results show that:

- The first **10 components** explain approximately **60.4%** of the total variance.
- The first **15 components** explain approximately **69.6%** of the total variance.
- The first **20 components** explain approximately **76.6%** of the total variance.

In this study, the first **15 principal components (PC1–PC15)** were selected, as they capture nearly **70%** of the total variance while substantially reducing the dimensionality of the original feature space. This provides a good balance between information preservation and model simplicity.

Relationship Between Components and Original Variables

Analysis of the PCA loading matrix indicates that each principal component represents a combination of several related socio-economic indicators:

- **PC1** is mainly influenced by labor and economic structure variables, such as employment characteristics and sectoral composition, reflecting the overall labor structure of a country.
- **PC2 and PC3** show high loadings on variables related to production, trade, and natural resources, capturing differences in productive capacity and resource utilization.
- **PC4–PC6** are associated with industrialization and trade intensity, including high-technology exports and merchandise trade indicators, representing levels of economic and technological development.

- **PC7–PC15** explain more specific variations related to employment composition, import–export structure, and trade specialization, capturing secondary but meaningful patterns in the data.

Overall, the selected principal components effectively summarize key socio-economic dimensions—labor structure, production, trade, resources, and technology—while preserving the majority of the information contained in the original variables.

2. **Method LDA**

In addition to PCA, Linear Discriminant Analysis (LDA) was applied as a supervised feature extraction method. Unlike PCA, which maximizes overall variance, LDA aims to find linear combinations of the original variables that best separate the predefined classes, in this case the Gender Ratio Class.

Number of Discriminants Selected

Since Gender Ratio Class consists of multiple classes, LDA can generate at most $(C - 1)$ discriminant components, where C is the number of classes. In this analysis, the resulting discriminant components were retained and used to represent the data, as they maximize the between-class variance while minimizing the within-class variance.

Relationship Between Discriminants and Original Variables

Based on the LDA coefficient matrix, the discriminant components are strongly influenced by socio-economic variables related to employment structure, trade, and technology. Variables such as vulnerable employment, fuel exports, high-technology exports, merchandise trade, and contributing family workers show relatively large coefficients, indicating their importance in distinguishing different Gender Ratio Classes.

These results suggest that differences in labor market vulnerability, trade composition, and technological development play a key role in separating countries into different gender ratio categories

Part 2.0 Data Cleaning and Preparation

Datasets B and C were merged using country names to combine socio-economic indicators with regional information. Prior to analysis, missing values and data structure were examined.

Since the data span multiple years, the dataset was **aggregated at the country level** by computing the **median values** of numerical indicators for each country. This aggregation reduces temporal noise and provides a stable representation of each country's socio-economic characteristics.

Countries with missing values in the target variable, **GNI per capita (PPP)**, were removed. Numerical variables that contained only missing values across all countries were also excluded from the analysis.

For the remaining numerical features with missing values, **median imputation** was applied using a SimpleImputer. Median imputation was chosen to minimize the influence of extreme values and maintain robustness against outliers.

The distribution of the target variable was then examined. **GNI per capita exhibits strong right-skewness**, and therefore a **log transformation** was applied prior to clustering to improve scale balance and interpretability.

Finally, the transformed target variable was **standardized** using StandardScaler before applying clustering, ensuring that distance-based methods were not dominated by scale differences.

Part 2.1 Clustering Analysis

Clustering Method and Performance Evaluation

In this analysis, **K-means clustering** was applied to group countries based on their economic characteristics, with **GNI per capita, PPP (current international $)** as the primary variable of interest. Prior to clustering, the data were aggregated at the country level, log-transformed, and standardized to ensure comparability and suitability for distance-based methods.

To evaluate clustering quality, the **silhouette score** was used. The final clustering solution achieved a silhouette score of **0.578**, indicating a reasonably good separation between clusters and meaningful internal cohesion within clusters.

Number of Clusters and Cluster Determination

The final number of clusters was set to **three (k = 3)**. This choice balances interpretability and clustering performance, allowing countries to be grouped into low-, middle-, and high-income categories based on their GNI per capita. The silhouette score further supports that this clustering structure provides clear separation among the clusters.

Characteristics of Countries in Each Cluster

Summary statistics of GNI per capita within each cluster reveal distinct economic profiles:

Cluster 0: Low-Income Countries

- Mean GNI per capita: approximately **3,398**
- GNI range: **810 – 6,050**
- Dominated by countries from **Africa**, with smaller representations from South Asia and Southeast Asia

Countries in this cluster share characteristics of **low income levels and limited economic resources**, reflecting developing economies with lower purchasing power.

Cluster 2: Middle-Income Countries

- Mean GNI per capita: approximately **13,386**
- GNI range: **6,270 – 23,210**
- Geographically diverse, including **Africa, Eastern Europe, North America, South America, and Asia**

This cluster represents **emerging and middle-income economies**, characterized by moderate income levels and transitional economic structures.

Cluster 1: High-Income Countries

- Mean GNI per capita: approximately **47,013**
- GNI range: **23,970 – 109,320**
- Predominantly composed of countries from **Western Europe, Northern Europe, and North America**

Countries in this cluster share characteristics of **high income levels, advanced economic development, and strong purchasing power**.

Comparison Between Clusters

Countries within the same cluster exhibit similar income levels and economic capacity, while clear differences exist between clusters:

- **Cluster 0 vs. Cluster 2:** Middle-income countries show substantially higher income levels and greater economic capacity compared to low-income countries.
- **Cluster 2 vs. Cluster 1:** High-income countries exhibit significantly higher GNI per capita and more advanced economic structures than middle-income countries.
- **Cluster 0 vs. Cluster 1:** The contrast between low- and high-income countries is pronounced, highlighting global economic inequality.

Part 2.2 Modeling 2: GNI per Capita Prediction

Workflow 1: StandardScaler + Linear Regression

Preprocessing Choices

In the first workflow, all numerical features were standardized using **StandardScaler**. This preprocessing step was chosen to ensure that variables measured on different scales contribute equally to the regression model. Standardization improves numerical stability and is essential for models that estimate coefficients directly.

Feature Subset Selection

All selected socio-economic features were retained in this workflow without explicit feature reduction. This approach serves as a **baseline model**, allowing the effect of using the full feature set to be evaluated before applying regularization or feature selection techniques.

Modeling Choice

A standard **linear regression model** was used due to its simplicity and interpretability. This model provides a clear baseline for understanding linear relationships between socio-economic indicators and GNI per capita.

Performance

- **RMSE:** 0.419
- **MAPE:** 0.036
- **R-squared:** 0.756

The results indicate that the model explains a large proportion of the variance in GNI per capita. However, the relatively higher RMSE and MAPE suggest that prediction error remains noticeable, potentially due to redundant or less informative features.

Workflow 2: Lasso Regression (Final Workflow)

Preprocessing Choices

As in Workflow 1, feature standardization was applied. This step is particularly important for **Lasso regression**, as the regularization penalty depends on the magnitude of feature coefficients.

Feature Subset Selection

Lasso regression performs **automatic feature selection** by shrinking less important coefficients toward zero. This effectively removes redundant and weak predictors, resulting in a more compact and informative feature set. Feature reduction helps reduce overfitting and improves model generalization.

Modeling Choice

Lasso regression was selected to address multicollinearity and control model complexity. By combining regression and feature selection, Lasso produces a more robust and interpretable model compared to the baseline linear regression.

Performance

- **RMSE:** 0.346
- **MAPE:** 0.029
- **R-squared:** 0.834

Compared to Workflow 1, Workflow 2 achieves **lower prediction error** and **higher explanatory power**, demonstrating improved overall performance.

Final Model Performance and Justification

- The final model (Workflow 2) clearly outperforms the baseline model. RMSE decreases from **0.419 to 0.346**, and MAPE decreases from **0.036 to 0.029**, indicating more accurate predictions both in absolute and relative terms. In addition, R-squared increases from **0.756 to 0.834**, showing that the model explains a substantially larger proportion of the variance in GNI per capita.
- Overall, the performance of the final model is considered **good** for cross-country socio-economic regression, as it balances prediction accuracy, robustness, and interpretability.
-

The final model explains a substantial portion of the variance in GNI per capita, indicating good predictive performance for cross-country economic data.

Model Interpretation

Analysis of the Lasso regression coefficients shows that variables related to **education attainment**, **labor force participation**, and **economic structure** have the strongest influence on GNI per capita. Education-related indicators exhibit a positive relationship with income levels, while labor market characteristics also play a significant role in explaining economic differences across countries.

By eliminating less informative variables, the final model highlights the most important socio-economic factors driving GNI per capita and provides clearer insights into the determinants of economic development.

Part 2.3 Dashboard and Narrative Visualization

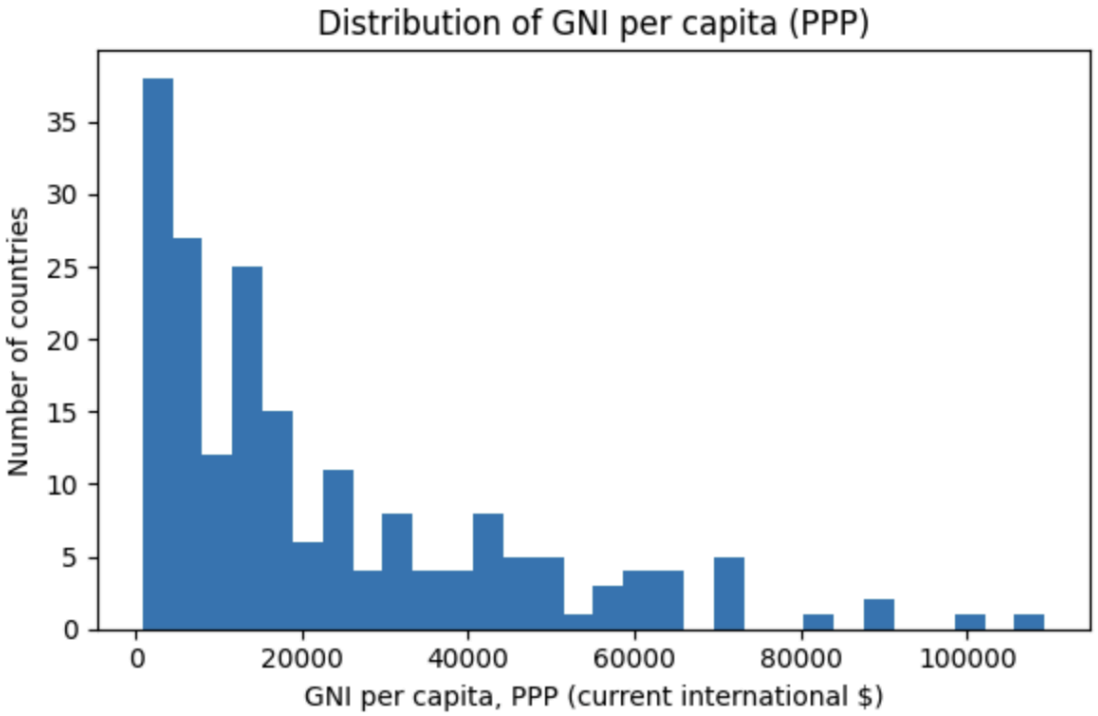
Objective of the Dashboard

The objective of this dashboard is to visually communicate global patterns and relationships between **economic development (GNI per capita)**, **education (tertiary school enrollment)**, and **regional distribution of countries**. The dashboard aims to provide both an overall global perspective and more detailed insights into how education and income are related across countries and over time.

1. Distribution of GNI per Capita (Histogram)

The histogram illustrates the distribution of **GNI per capita (PPP)** across countries.

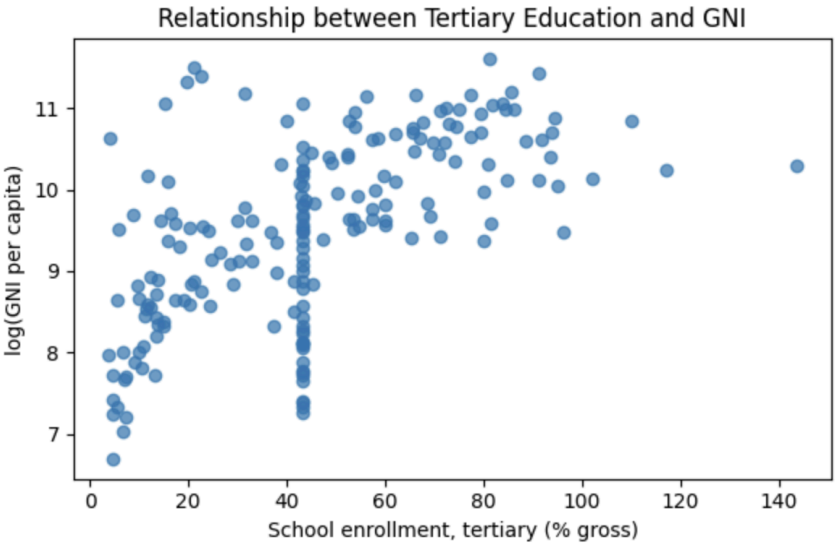
The objective of this graph is to highlight the strong right-skewness in income distribution, where most countries fall into lower income ranges while a small number of countries have very high income levels. This visualization motivates the use of **log transformation** in subsequent analyses.



2. Relationship between Tertiary Education and GNI (Scatter Plot)

The scatter plot shows the relationship between **tertiary school enrollment (% gross)** and **log-transformed GNI per capita**.

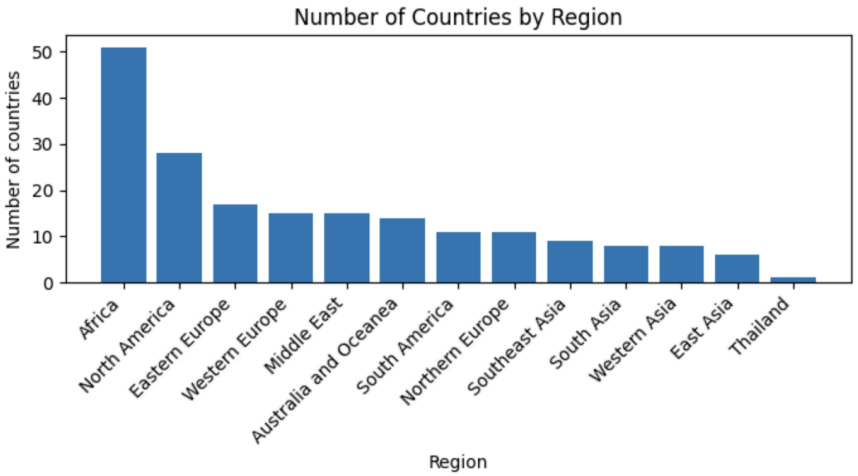
The objective of this graph is to explore the association between higher education attainment and economic prosperity. The upward trend observed in the plot suggests a positive relationship between tertiary education and income level, supporting the regression analysis conducted in Part 2.2.



3. Number of Countries by Region (Bar Chart)

This bar chart displays the number of countries in each **region** included in the dataset.

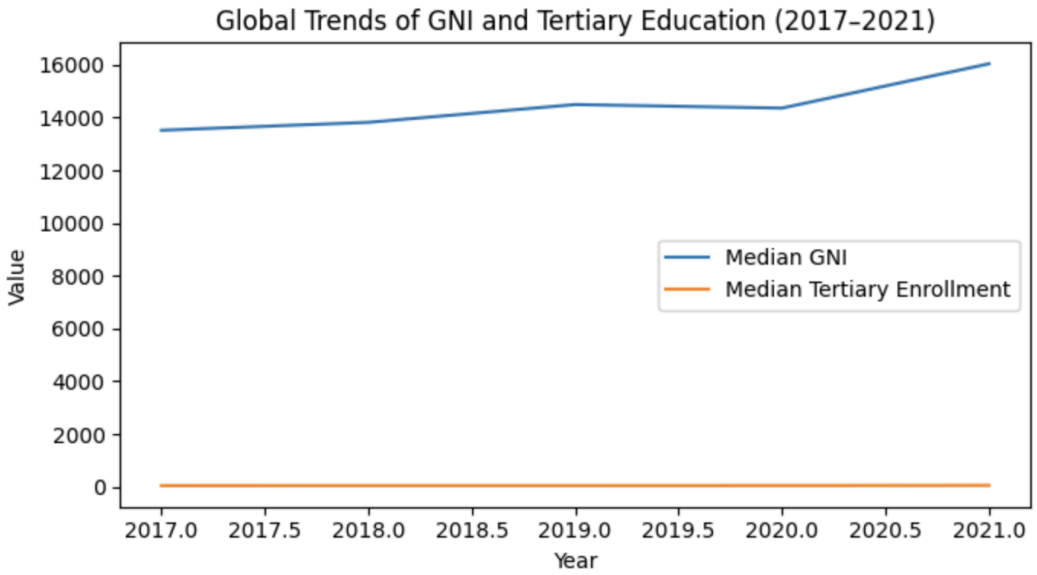
The objective of this visualization is to provide context about the regional composition of the data and to highlight potential regional imbalances. Understanding this distribution is important for interpreting clustering and regression results, as some regions are more heavily represented than others.



4. Global Trends of GNI and Tertiary Education (Time-Series Plot)

The time-series plot presents the **median GNI per capita** and **median tertiary school enrollment** from **2017 to 2021**.

The objective of this graph is to illustrate global trends over time, showing how income levels and higher education participation evolve together. This visualization provides a temporal perspective that complements the cross-sectional analyses in earlier sections.



Narrative Visual Structure

The dashboard follows a **martini-glass narrative structure**. It begins with broad, descriptive visualizations that provide a global overview (distribution of GNI and regional composition). It then narrows the focus to analytical insights through relationship and trend visualizations (education–income relationship and time-series trends). This structure guides the viewer from high-level context to more detailed interpretation, enabling a clear and coherent data-driven story.